Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – III (Batch 2024-2028)

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 3	BSC	24BT201T	Biophysics & Structural Biology	3	1	0	4	4
	ESC	24BT202T	Introduction to Artificial Intelligence	3	0	0	3	3
	PC	24BT203T	Principles of Biochemistry	3	0	0	3	3
	PC	24BT203P	Biochemistry Lab	0	0	2	2	1
	PC	24BT204T	Human Anatomy and Physiology - I	3	0	0	3	3
	PC	24BT204P	Human Anatomy and Physiology – I Laboratory	0	0	2	2	1
	PC	24BT205T	General Microbiology	3	0	0	3	3
	PC	24BT205P	General Microbiology Laboratory	0	0	2	2	1
	Pro	24INT151	Civic and Social Service Internship	0	0	0	0	1

24BT201T					Biophysics and Structural Biology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. Learn the strategy and tactics of biophysical concepts of macromolecules and the conformational analysis
2. Understand the kinetics to study the ligand interaction with macromolecules.
3. Gain the fundamental on forces those determine the protein and nucleic acid structure and ligand interaction with macromolecules.
4. Learn the concepts of size and shape of the macro molecule using different techniques using various tools like X-ray crystallography and other techniques is explained.

UNIT I: INTRODUCTION 10 Hrs.

Levels of structures in biological macromolecules, basic strategies in biophysics, Principles and concepts used in biophysical analysis of life processes, biomolecules and their interactions, size and shape of macromolecules.

UNIT II: CONFORMATIONAL ANALYSIS OF PROTEINS 10 Hrs.

Basic problems, polypeptide chains geometrics, potential energy calculations, observed values for rotation angles, hydrogen bonding, hydrophobic interactions and water structures, ionic interactions and disulphide bonds, Prediction of proteins structure.

UNIT III: CONFORMATIONAL ANALYSIS OF NUCLEIC ACIDS 10 Hrs.

General characteristics of nucleic acid structure, geometrics, glycosidic bond, rotational isomers and those puckering, backbone rotational angles and steric hindrances, forces stabilizing ordered forms, base pairing, base stacking, tertiary structure of nucleic acids.

UNIT IV: TECHNIQUES FOR THE STUDY OF BIOLOGICAL STRUCTURE AND FUNCTION 10 Hrs.

Size and shape of macromolecules, methods of direct visualisation, macromolecules as hydrodynamic particles, macromolecular diffusion, ultracentrifugation, viscometer, X-ray crystallography: X–ray diffraction, determination of molecular structures, electron microscopy, neutron scattering, light scattering.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify and Compare the different macromolecules
CO2 : Gain the fundamental knowledge on conformational analysis of biomolecules
CO3 : Understand the principles of biological structure and their functions
CO4 : Get acquainted with the techniques for the studying macromolecules
CO5 : Focus professionally on analytical techniques and methods
CO6 : Develop solution of different potential energy calculations and prediction of protein structure.

TEXT/REFERENCE BOOKS

1. Cantor, R. and Schimmel, P. R., "Biophysical Chemistry", Vol. I, II, W. H. Freeman & Co., 2018.
2. Gu, J. and Bourne, P. E., "Structural Bioinformatics", 2nd Edition, Wiley, 2019.
3. I.N. Levine, "Quantum Chemistry", (2020), Pearson Educ., Inc., New Delhi.

24BT202T					Introduction to Artificial Intelligence					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25			100

COURSE OBJECTIVES

1. Understand the basic concepts of Artificial Intelligence
2. Learn the applications of artificial intelligence in bioinformatics
3. Learn some basic search algorithms for problem solving; knowledge representation and reasoning; pattern recognition; fuzzy logic; and neural networks.
4. Understand the advancements of designing the intelligent systems that can solve general purpose problems, represent and process knowledge, plan and act, reason under uncertainty.

UNIT I: INTRODUCTION 10 Hrs.

Introduction to Artificial Intelligence (AI): Introduction to Artificial Intelligence, Problems, Approaches and tools for Artificial Intelligence. Introduction to search, Search algorithms, Heuristic search methods, Optimal search strategies. Use of graphs in Bioinformatics. Current Techniques of Artificial Intelligence: Introduction to probability, Bayes' theorem, Bayesian networks and Markov networks.

UNIT II: CLASSIFICATION METHODS 10 Hrs.

Linear Classifiers & Logistic Regression ,Linear Classifiers, Overfitting & Regularization in Logistic Regression, Decision Trees, Preventing Overfitting in Decision Trees, Handling Missing Data, Clustering and retrieval of data, Nearest Neighbour Search, Clustering with k-means, Hierarchical Clustering.

UNIT III: SUPERVISED/ UNSUPERVISED LEARNING 10 Hrs.

parametric/non-parametric algorithms, support vector machines, kernels, neural networks, Unsupervised learning - clustering, dimensionality reduction, recommender systems, deep learning, Best practices in machine learning (bias/variance theory; innovation process in machine learning and AI, Support vector machines (SVMs), case studies and applications..

UNIT IV: APPLICATIONS OF AI IN BIOTECHNOLOGY 10 Hrs.

Genetic programming and Bioinformatics applications. Bayesian Networks and Fuzzy Neural Networks with case studies. Decision trees –Statistical Learning methods - Introduction to deep learning. Syntactic processing, semantic processing, Gesture recognition, face recognition, speech recognition.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Learn about concepts of artificial intelligence and their applications in Bioinformatics
CO2 : Use the knowledge acquired for both problem solving and for reasoning
CO3 : Understand the basic ideas and techniques underlying the design of intelligent computer Systems
CO4 : Use the Techniques and algorithms to address those problems.
CO5 : Focus on problems, the ethical, legal and social issues involved in the field of AI
CO6 : Develop solution AI application in health and biotechnology sector.

TEXT/REFERENCE BOOKS

1. K. G. Srinivasa, G. M. Siddesh, S. R. Manisekhar, "Statistical Modelling and Machine Learning Principles for Bioinformatics Techniques, Tools, and Applications", published by Springer Nature, 2020. ISBN: 9789811524455.
2. Stuart Jonathan Russell and Peter Norvig "Artificial Intelligence: A Modern Approach", Prentice Hall, 2016. ISBN 9781537600314
3. Zheng Rong Yang , "Machine Learning Approaches to Bioinformatics", World Scientific Publishing Co. Pte. Ltd, 2010.ISBN 981-4287-30-X.
4. Vincent François-Lavet, Peter Henderson, Riashat Islam, "An Introduction to Deep Reinforcement Learning", Now Publishers, 2019. ISBN: 9781680835380.

24BT203T					Principles of Biochemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. Develop knowledge of the fundamental chemical principles that govern biological systems
2. Distinguish biomolecules and their metabolic pathways
3. Discover the comprehension of the regulation of biological/biochemical processes
4. Develop an understanding of the metabolic pathways of biomolecules and their regulation in biological systems.

UNIT I: INTRODUCTION 10 Hrs.

Introduction to biological system, Review on cellular, chemical, physical, genetic and evolutionary backgrounds to biochemistry; Properties of water- solubility, ionization property and water as a reactant, pH and buffers and their importance.

UNIT II: AMINO ACIDS AND PROTEINS 10 Hrs.

Amino acids and Proteins: Classification, structure and biological importance of amino acids, acid–base chemistry and stereochemistry of amino acids. Proteins: Primary, secondary, tertiary, quaternary & domain structure of proteins. Protein denaturation and renaturation. Enzyme and its classification.

UNIT III: CARBOHYDRATES 10 Hrs.

Carbohydrates: Definitions, biological functions, Physical properties. Classification of carbohydrates. Metabolism of carbohydrates: Glycolysis, TCA cycle, oxidative phosphorylation, gluconeogenesis and pentose phosphate pathway and their regulation.

UNIT IV: NUCLEIC ACIDS AND LIPIDS 10 Hrs.

Nucleic acids: Composition, properties and function. Metabolism- synthesis of purines and pyrimidine. Lipids: Classification, structure, properties, function and metabolism of fatty acids. Classification, structure, properties and biological function of simple lipids – triacylglycerol and waxes. Compound lipids- phospholipids and glycolipids. Cholesterol- structure, properties and importance.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the fundamentals of biomolecules.
CO2 : Explain the importance of biological buffer systems.
CO3 : Classify amino acids and proteins and describe their functions.
CO4 : Analyze the metabolic pathways of carbohydrates.
CO5 : Evaluate the relationship between the structure and function of nucleic acids.
CO6 : Create a model of lipid functions in biological systems.

TEXT/REFERENCE BOOKS

1. David L. Nelson and Michael M. Cox. 2020," Lehninger Principles of Biochemistry: International Edition", 7th edition, W.H. Freeman. USA.
2. Berg, Tymoczko and Stryer, "Biochemistry" (5th Edition, 2007) W.H. Freeman Press, SanFrancisco, USA.

24BT203P					Principles of Biochemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To learn the basics of biochemistry.
2. To relate concepts learned in chemistry and engineering to real-world situations.
3. To acquire skills to perform hematological assays.
4. To demonstrate various enzymatic activities.

LIST OF EXPERIMENTS

1 Laboratory practices in biochemistry and reagent preparation (calculations)
2 Qualitative analysis of carbohydrates (Glucose, Fructose, Lactose, Maltose, Sucrose andstarch)
3 Identification tests for Proteins (albumin and Casein)
4 Quantitative analysis of reducing sugars (DNSA method) and Proteins (Biuret method)
5 Qualitative analysis of urine for abnormal constituents
6 Determination of blood creatinine
7 Determination of blood sugar
8 Determination of serum total cholesterol
9 Preparation of buffer solution and measurement of pH
10 Study of enzymatic hydrolysis of starch
11 Determination of Salivary amylase activity
12 Study the effect of Temperature on Salivary amylase activity.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Apply Laboratory Fundamentals
CO2 : Analyze Biomolecules
CO3 : Interpret Diagnostic Values
CO4 : Understand Buffer Systems
CO5 : Investigate hematology parameters
CO6 : Evaluate Enzyme Function

TEXT/REFERENCE BOOKS

1. Lehninger, “Principles of Biochemistry”, W H Freeman & Co; 8th edition.
2. Robert K. Murry, Daryl K. Granner and Victor W. Rodwell ,”Harper’s Biochemistry”, McGraw-Hill Education.
3. D. Satyanarayan and U.Chakrapani ,”Biochemistry”, Elsevier
4. Conn and Stump, “Outlines of Biochemistry”, New York : Wiley

24BT204T					Human Anatomy and physiology -I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Gain fundamental knowledge on Structure and functions of the various organs of the human body
- Learn the homeostatic mechanisms and their imbalances in the human body
- Understand the different Various vital physiological parameters of the human body and their clinical significance.
- Understand the coordinated working pattern of different organs of each system

UNIT I: Cell and Tissues of Human Body — 10 Hrs

Scope of Anatomy and Physiology, Structure of Cell: Components and its functions. Classification on Tissues of the human body, structure, location and functions of epithelial, muscular and nervous and connective tissues

UNIT II: Blood and Lymphatic system — 10 Hrs

Haemopoietic system: Composition and functions of blood, Characteristics and functions of Blood cells, Mechanism of Blood Clotting, Importance of Blood grouping. Lymphatic organs and tissues, lymphatic vessels, lymph circulation and functions of lymphatic system

UNIT III: Respiratory system & Cardiovascular system — 10 Hrs

Anatomy of respiratory organs and their functions, Regulation, and Mechanism of respiration. Respiratory volumes and capacities. Anatomy and Physiology of heart. Blood vessels and circulation. Cardiac cycle, Blood pressure and its regulation

UNIT IV: Nervous system and Sense organs — 12 Hrs

Classification of nervous system. Classification of nervous system. Anatomy and physiology of cerebrum, cerebellum, mid brain. Function of hypothalamus, medulla oblongata and basal ganglia. Spinal cord-structure and reflexes. Names and functions of cranial nerves. Sense organs: Eye, Ear, Skin,Tongue and Nose

Max. 42 Hr.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Gain fundamental knowledge of human anatomy and physiology.
CO2: Understand the functions of Cell, tissues, muscles, vital organs and their control mechanism.
CO3: Classify and compare various blood cells, respiratory mechanisms, cardiac rhythm, nerves, organs and their applications.
CO4: Get acquainted with homeostatic mechanisms of human body.
CO5: Focus on consequence in organ dysfunction and its clinical significance.
CO6: Design and develop solutions to various clinical conditions.

TEXT/REFERENCE BOOKS

1. Ross and Wilson Anatomy and Physiology in Health and illness
2. Fundamentals of Medical Physiology by K. Sambulingam and P Sambulingam
3. Human Anatomy and Physiology by S. Chaudhary and A. Chaudhary
4. Human Physiology by C. C. Chatterjee
5. Goyal R.K., Natvar M.P. and Shah S.A., Practical Anatomy, Physiology and Biochemistry, Experimental Physiology

24BT204P					Human Anatomy and physiology Lab -I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. To introduce the fundamentals of human anatomy and physiology through practical experiments.
2. To develop skills in using laboratory equipment and performing hematological assessments.
3. To understand and identify the structure and functions of bones, tissues, and organs.
4. To learn vital physiological measurements and their clinical relevance.

List of Experiments

1. Study of compound microscope.
2. Microscopic study of Tissues
3. Identification of axial bones
4. Identification of appendicular bones
5. Identification of organs with charts, models and slides
6. Introduction to Equipment's.
7. Determination of heart rate
8. Determination of pulse rate
9. Determination of Body temperature.
10. Recording of blood pressure.
11. Determination of bleeding time
12. Determination of clotting time

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Demonstrate basic knowledge of human anatomy and physiology through hands-on experiments.

CO2: Identify and describe axial and appendicular bones using charts, models, and specimens.

CO3: Perform hematological tests, including the determination of bleeding and clotting time.

CO4: Accurately measure and interpret physiological parameters like heart rate, pulse rate, body temperature, and blood pressure.

CO5: Use laboratory equipment effectively for studying tissues and physiological processes.

CO6: Apply standard laboratory protocols to analyze physiological data and ensure accuracy in experimentation.

REFERENCE BOOKS

1.Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill Livingstone, New York

2. Textbook of Human Histology by Inderbir Singh, Jaypee brother's medical publishers, New Delhi.

3. Textbook of Practical Physiology by C.L. Ghai, Jaypee brother's medical publishers, New Delhi.

4. Practical workbook of Human Physiology by K. Srinageswari and Rajeev Sharma, Jaypee brother's medical publishers, New Delhi.

24BT205T					General Microbiology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. Apply fundamental microbiology principles, including classification, diversity analysis, microscopy, staining, and aseptic technique.
2. Demonstrate proficiency in core laboratory techniques for isolating, culturing, and identifying microorganisms.
3. Analyze microscopic structures, staining results, and molecular data to classify microorganisms and study their cellular architecture.
4. Explain the principles underlying sterilization, media preparation, and cell growth, and understand the membrane transport mechanisms.

UNIT I: HISTORY AND CLASSIFICATION 10 Hrs.

Concepts of species and hierarchical taxa. Bergey. 's system of classification. Classification of Bacteria, Fungi, and Viruses; Modern methods to study microbial diversity: NGS. MiSeq; Molecular Taxonomy- 16S rRNA gene sequencing, Phylogenetic grouping. Fatty Acid Methyl Ester (FAME) analysis, ITS; Methods to study microbial community: DGGE, SSCP, T-RFLP.

UNIT II: MICROSCOPY AND STAINING 10 Hrs.

Amino acids and Proteins: Classification, structure and biological importance of amino acids, acid–base chemistry and stereochemistry of amino acids. Proteins: Primary, secondary, tertiary, quaternary & domain structure of proteins. Protein denaturation and renaturation. Enzyme and its classification.

UNIT III: SUBCELLULAR STRUCTURES 10 Hrs.

Microbial cell: Ultrastructure of bacteria, subcellular structures and cell envelope – slime, capsule, cell wall, pili, flagella, cell inclusions, biosynthesis of bacterial cell wall, cell membrane – Bio-membrane, liposomes – membrane transport – diffusion, active and passive transport and osmoregulation.

UNIT IV: STERILIZATION & MEDIA AND TECHNIQUES 10 Hrs.

Sterilization and Disinfection- principles -methods of sterilization- physical methods - Dry heat- Moistheat- Radiation. Filtration (Membrane and HEPA). Chemical sterilization- Chemical agents- mode of action- phenol coefficient test sterility testing. Culture and media preparation - solid and liquid. Types of media- Semi synthetic, Synthetic, Enriched, Enrichment, Selective and Differential media. Pure culture techniques – Tube dilution, Pour, Spread, Streak plate. Anaerobic culture technique Wright's Tube, Roll tube method, Anaerobic Jar.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Apply fundamental principles of microbiology, including classification, diversity analysis, microscopy, staining, and aseptic technique.

CO2 : Demonstrate proficiency in core laboratory techniques for isolating, culturing, and identifying microorganisms.

CO3 : Analyze microscopic structures, staining results, and biochemical data to classify microorganisms and study their cellular architecture.

CO4 : Explain the principles underlying sterilization, media preparation, and the mechanisms of cell growth and transport

CO5 : Utilize modern molecular tools (e.g., 16S rRNA sequencing, FAME) and community analysis techniques to study microbial diversity and interactions.

CO6 : Apply critical thinking and scientific reasoning to design microbiological experiments, interpret results, and evaluate techniques.

TEXT/REFERENCE BOOKS

1. Prescott, Harley, Klein. 2003," Microbiology",5th Edition. McGraw Hill Publ.
2. Bernard R. Glick & Jack J. Pasternak. 2002," Molecular Biotechnology",Indian edition. Panima Publishing Corporation.
3. M. Madigan, K. Bender, D. Buckley, W. Sattley, D. Stahl, "Brock Biology of Microorganisms", 15th Edition. Pearson Education. 2018.
4. Tortora, G.J., Funke, B.R. and Case, C.L. 2012,"Microbiology - An Introduction" 11th Edition. Pearson Education.

24BT205T					General Microbiology Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Perform Microbial Isolation and Cultivation
2. Analyze Microbial Morphology and Physiology
3. Explore Fungal Identification Techniques
4. Assess Microbiological Quality of Food and Water

LIST OF EXPERIMENTS

1 Techniques for isolation of microorganisms-Pour plate technique and Spread plate technique
2 Purification of microorganisms from a mixed culture by streak plate preparation
3 Preservation of microorganisms-Glycerol stock method and Agar slant method
4 Microscopy: Microscopic examination of stained cell preparation-Preparation of bacterial smears
5 Simple staining and gram staining
6 Microscopic examination of live bacterial population-Hanging drop method
7 Biochemical Activities of Microorganisms: IMViC, catalase, oxidase
8 The Fungi Cultivation and identification of unknown fungi by lactophenol cotton blue staining
9 Microbiology of food-Methylene Blue Reductase Test and Microbiological Analysis of Food products
10 Microbiology of water Standard Qualitative analysis of water: Presumptive test, Confirmedtest, Completed Test

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Prepare basic solutions required for complex analysis.
CO2 : Employ tools in biochemistry to assess biomolecules.
CO3 : Discuss the various steps of analysis.
CO4 : Interpret the results obtained from assay methods
CO5 : Develop different strategies for the analysis of novel molecules.
CO6 : Criticize the results and validate methods.

TEXT/REFERENCE BOOKS

1. Melvyn Kay, "**Practical Handbook of Microbiology** "(2nd edition), CRC Press, 2008.
2. M. Madigan, K. Bender, D. Buckley, W. Sattley, D. Stahl, **"Brock Biology of Microorganisms"**, 15th Edition. Pearson Education. 2018.